Identifying EEZ Regions Suitable for Marine Aquaculture

Oliver Witt

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Background and instructions

Determining which Exclusive Economic Zones (EEZ) on the West Coast of the US are best suited to developing marine aquaculture for several species of oysters and a species of your choice. Suitable locations will be determined based on range of suitable sea surface temperature (SST) and depth values for the species.

Choose a species

- Red Abalone (https://www.sealifebase.ca/summary/Haliotis-rufescens.html)
- Depth: 0-24m
- Temperature: 8°C - 18°C

Workflow outline

Prepare data

```
# Load the packages we might need
library(tidyverse)
library(sf)
library(terra)
library(tmap)
library(kableExtra)
library(spData)
```

```
library(spDataLarge)
library(geodata)
library(elevatr)
library(tigris)
library(here)
getwd()
```

```
[1] "/Users/oMac/Desktop/meds_stuff/eds223 repos folder/eds223-homeworks/EEZs-for-Marine-Aquaculture/EEZs-for-Marine-Aquaculture"
```

```
# Read in the data
# SST files (rasters)
sst_2008 <- terra::rast(here("data", "average_annual_sst_2008.tif"))
sst_2009 <- terra::rast(here("data", "average_annual_sst_2009.tif"))
sst_2010 <- terra::rast(here("data", "average_annual_sst_2010.tif"))
sst_2011 <- terra::rast(here("data", "average_annual_sst_2011.tif"))
sst_2012 <- terra::rast(here("data", "average_annual_sst_2012.tif"))
# Depth file (raster)
depth <- terra::rast(here("data", "depth.tif"))
# EEZ file (shapefile)
wc_eez_sf <- read_sf(here("data", "wc_regions_clean.shp"))

# Combine SST rasters into a raster stack
sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)

# Check the class of the new raster (sanity check)
class(sst_stack)
```

```
[1] "SpatRaster"
attr(,"package")
[1] "terra"
```

/Users/oMac/Desktop/meds_stuff/eds223 repos folder/eds223-homeworks/EEZs-for-Marine-Aquaculture/EEZs-for-Marine-Aquaculture/aquaculture_analysis.qmd ### Process data

```
# Find mean SST
sst_mean <- mean(sst_stack, na.rm=TRUE)
sst_mean
```

```
class       : SpatRaster
size        : 480, 408, 1  (nrow, ncol, nlyr)
resolution  : 0.04166185, 0.04165702  (x, y)
extent      : -131.9848, -114.9867, 29.99305, 49.98842  (xmin, xmax, ymin, ymax)
coord. ref. : lon/lat WGS 84
source(s)   : memory
name        :    mean
min value   : 278.130
max value   : 306.045
```

```
# Convert avg SST from K to C
sst_mean_c <- sst_mean - 273.15

# Crop depth raster to match extent of SST raster
depth_crop <- crop(depth, sst_mean_c)

# Resample depth to match resolution of SST data using  bilinear method approach
depth_resampled <- resample(depth_crop, sst_mean_c, method = "bilinear")

# Check that depth and SST match in resolution, extent, and CRS
crs(depth_resampled) == crs(sst_mean_c)
```

```
[1] FALSE
```

```
# They do not match in CRS, so we need to match them
depth_proj <- project(depth_crop, sst_mean_c)
depth_resampled <- resample(depth_proj, sst_mean_c, method = "bilinear")
crs(depth_resampled) == crs(sst_mean_c)
```

```
[1] TRUE
```

```
# They now match in CRS.  Check resolution and extent
ext(depth_resampled) == ext(sst_mean_c)
```

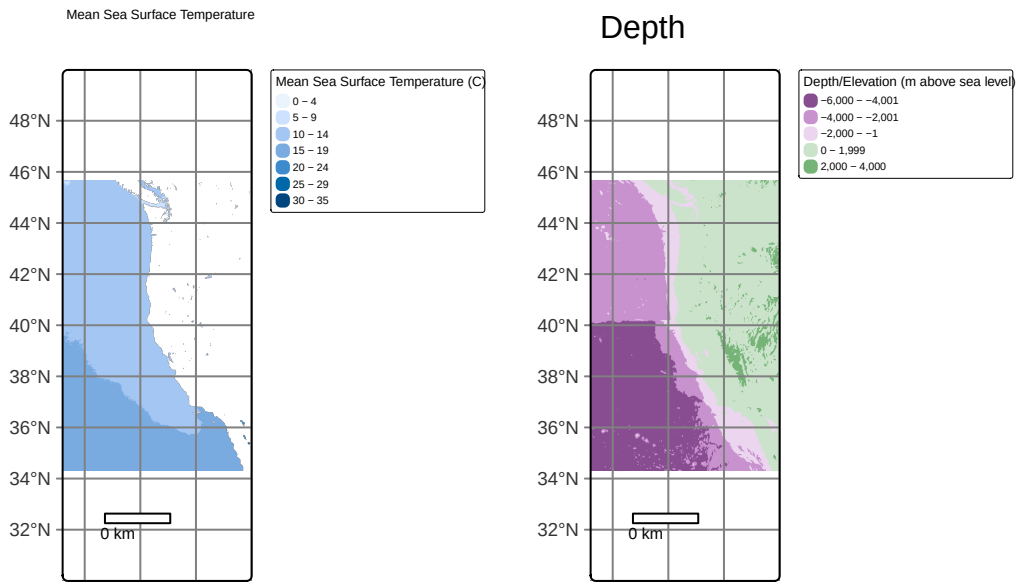
```
[1] TRUE
```

```
res(depth_resampled) == res(sst_mean_c)
```

```
[1] TRUE TRUE
```

```
# They match!

# Let's visualize what we have so far:
prelim_sst_map <- tm_shape(sst_mean_c) +
     tm_raster(col.legend = tm_legend(title = "Mean Sea Surface Temperature (C)")) +
     tm_title(text = "Mean Sea Surface Temperature", size = 1) +
     tm_graticules() +
     tm_scalebar(position = tm_pos_in("right", "bottom"))
prelim_depth_map <- tm_shape(depth_resampled) +
     tm_raster(col.legend = tm_legend(title = "Depth/Elevation (m above sea level)")) +
     tm_title(text = "Depth", size = 1) +
     tm_graticules() +
     tm_scalebar(position = tm_pos_in("right", "bottom"))
tmap_arrange(prelim_sst_map, prelim_depth_map, nrow = 1)
```



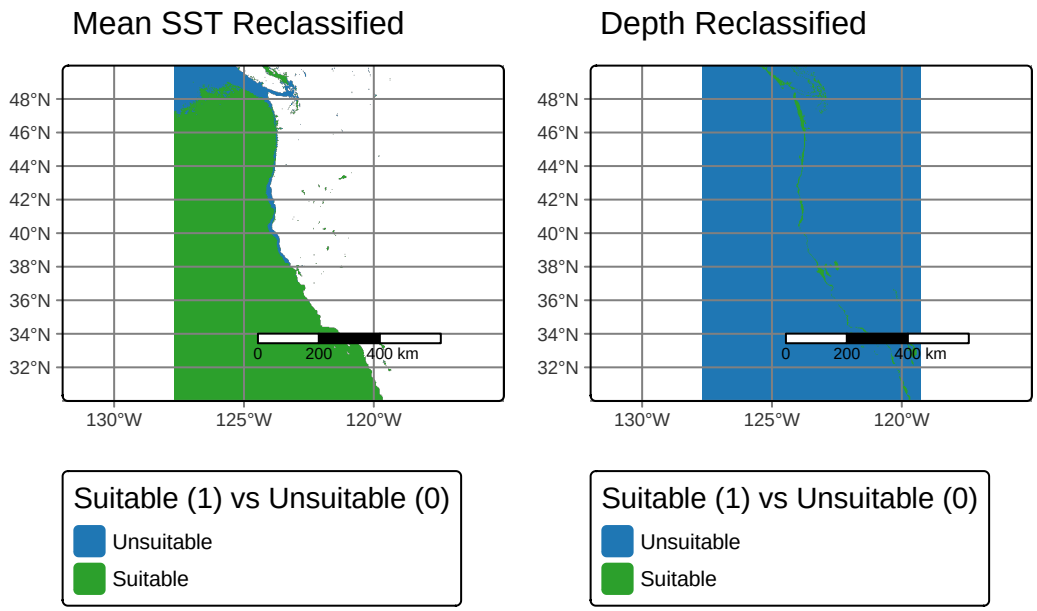
Find suitable locations

```
# Reclassify SST and depth data into locations that are suitable for oysters
# Set as "1" if depth is between 0-70m, otherwise set as zero
depth_matrix <- matrix(c(- Inf, -70, 0,
```

```
                          -70, 0, 1,
                          0, Inf, 0),
                          ncol = 3, byrow = TRUE)
depth_reclassified <- classify(depth_resampled, rcl = depth_matrix, include.lowest = TRUE)

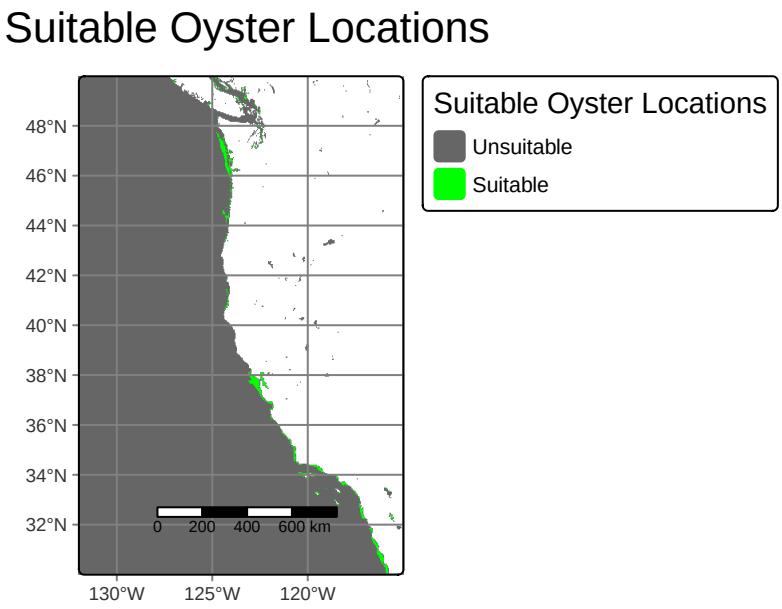
# Do the same thing for SST, setting to zero if temp is not 11-30C
sst_matrix <- matrix(c(-Inf, 11, 0,
                          11, 30, 1,
                          30, Inf, 0),
                          ncol = 3, byrow = TRUE)
sst_reclassified <- classify(sst_mean_c, rcl = sst_matrix, include.lowest = TRUE)

# Let's look at what we have now:
reclass_sst_map <- tm_shape(sst_reclassified) +
     tm_raster(col.scale = tm_scale_categorical(labels = c("Unsuitable", "Suitable")),
     col.legend = tm_legend(title = "Suitable (1) vs Unsuitable (0)")) +
     tm_title(text = "Mean SST Reclassified", size = 1) +
     tm_graticules() +
     tm_scalebar(position = tm_pos_in("right", "bottom")) +
     tm_options(component.autoscale = FALSE)
reclass_depth_map <- tm_shape(depth_reclassified) +
     tm_raster(col.scale = tm_scale_categorical(labels = c("Unsuitable", "Suitable")),
     col.legend = tm_legend(title = "Suitable (1) vs Unsuitable (0)")) +
     tm_title(text = "Depth Reclassified", size = 1) +
     tm_graticules() +
     tm_scalebar(position = tm_pos_in("right", "bottom")) +
     tm_options(component.autoscale = FALSE)
tmap_arrange(reclass_sst_map, reclass_depth_map, nrow = 1)
```




```
# Find locations that satisfy both SST and depth conditions, using "lapp"
suitable_oyster <- lapp(c(sst_reclassified, depth_reclassified), fun = function(x, y) x*y)

# Map what we have now (the areas that satisfy the depth and temp requirements for oysters)
satisfy_map <- tm_shape(suitable_oyster) +
     tm_raster(col.scale = tm_scale_categorical(labels = c("Unsuitable", "Suitable"), values
          col.legend = tm_legend(title = "Suitable Oyster Locations")) +
     tm_title(text = "Suitable Oyster Locations") +
     tm_graticules() +
     tm_scalebar(position = tm_pos_in("right", "bottom")) +
     tm_options(component.autoscale = FALSE)
satisfy_map
```



Determine the most suitable EEZs

```
# Select suitable cells within West Coast EEZs.  We will need to overlay the suitability map

# First, we need to make sure the CRS match
crs(wc_eez_sf) == crs(suitable_oyster)
```

```
[1] FALSE
```

```
# CRS do NOT match
wc_eez_sf_matched <- st_transform(wc_eez_sf, crs(suitable_oyster))
crs(wc_eez_sf_matched) == crs(suitable_oyster)
```

```
[1] TRUE
```

```
# Next, find area of grid cells
cell_area <- cellSize(suitable_oyster, unit = "km")
suitable_area <- cell_area * suitable_oyster
```

```r
# Rasterize wc_eez_sf, so that both EEZ data and suitable sites data are rasters
wc_eez_sf_matched$id <- seq_len(nrow(wc_eez_sf_matched))
eez_raster <- rasterize(wc_eez_sf_matched, suitable_oyster, field = "id")

# Find the total suitable area within each EEZ
area_in_eez <- zonal(suitable_area, eez_raster, fun="sum", na.rm = TRUE)
names(area_in_eez) <- c("id", "suitable_area_km2")

# Create a final dataset of EEZs by area that is suitable for oysters
eez_fin <- wc_eez_sf_matched |>
     mutate(id = row_number()) |>
     left_join(area_in_eez, by = "id") |>
     mutate(suitable_area_km2 = coalesce(suitable_area_km2, 0)) |>
     arrange(desc(suitable_area_km2)) |>
     mutate(rank = row_number())
eez_fin
```

```
Simple feature collection with 5 features and 8 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
# A tibble: 5 x 9
  rgn            rgn_key area_m2 rgn_id area_km2                 geometry    id
* <chr>          <chr>     <dbl>  <int>    <dbl>        <MULTIPOLYGON [°]> <dbl>
1 Central Calif~ CA-C    2.03e11      3  202738. (((-122.9928 38.28022, -
~     3
2 Southern Cali~ CA-S    2.07e11      4  206861. (((-120.6505 34.97431, -
~     4
3 Washington     WA      6.69e10      5   66898. (((-122.7675 49.00031, -
~     5
4 Oregon         OR      1.80e11      1  179994. (((-123.4318 46.23363, -
~     1
5 Northern Cali~ CA-N    1.64e11      2  164379. (((-124.2102 41.99843, -
~     2
# i 2 more variables: suitable_area_km2 <dbl>, rank <int>
```

Function

Wrap all of the code inside a function so that it is reproducible for any species, provided that we know the depth and temperature requirements for that species.

```r
# This version of the function is further simplified so that the only input is the species o

# Create pre-filled oyster and red abalone dataframes first (can do this easily for any spec
oyster <- data.frame(
     name = "Oyster",
     min_sst = 11,
     max_sst = 30,
     min_depth = 0,
     max_depth = 70
)
red_abalone <- data.frame(
     name = "Red Abalone",
     min_sst = 8,
     max_sst = 18,
     min_depth = 0,
     max_depth = 24
)

# Now create the function, where the sole argument is the dataframe for species of interest
simplified_aquaculture_analysis <- function(species){
     # read in data
     sst_2008 <- terra::rast(here("data", "average_annual_sst_2008.tif"))
     sst_2009 <- terra::rast(here("data", "average_annual_sst_2009.tif"))
     sst_2010 <- terra::rast(here("data", "average_annual_sst_2010.tif"))
     sst_2011 <- terra::rast(here("data", "average_annual_sst_2011.tif"))
     sst_2012 <- terra::rast(here("data", "average_annual_sst_2012.tif"))
     # Depth file (raster)
     depth <- terra::rast(here("data", "depth.tif"))
     # EEZ file (shapefile)
     wc_eez_sf <- read_sf(here("data", "wc_regions_clean.shp"))

     # Clean/crop/reshape data
     sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
     sst_mean_c <- mean(sst_stack, na.rm=TRUE) - 273.15
     depth_crop <- crop(depth, sst_mean_c)
     depth_proj <- project(depth_crop, sst_mean_c)
     depth_resampled <- resample(depth_proj, sst_mean_c, method = "bilinear")

     # Filter for suitable locations
     depth_matrix <- matrix(c(- Inf, -(species$max_depth), 0,
                              -(species$max_depth), -(species$min_depth), 1,
                              -(species$min_depth), Inf, 0),
```

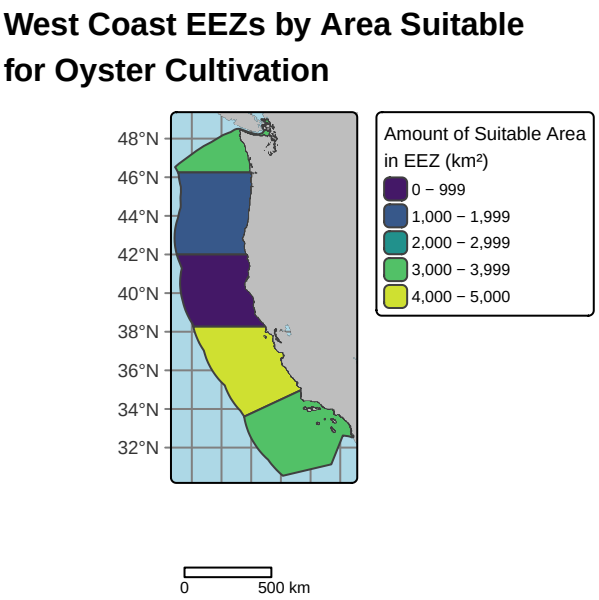
```
                          ncol = 3, byrow = TRUE)
depth_reclassified <- classify(depth_resampled, rcl = depth_matrix, include.lowest = TRU
sst_matrix <- matrix(c(-Inf, species$min_sst, 0,
                          species$min_sst, species$max_sst, 1,
                          species$max_sst, Inf, 0),
                          ncol = 3, byrow = TRUE)
sst_reclassified <- classify(sst_mean_c, rcl = sst_matrix, include.lowest = TRUE)
suitable_spots <- lapp(c(sst_reclassified, depth_reclassified), fun = function(x, y) x*y

# Calculate suitable area by EEZ
wc_eez_sf_matched <- st_transform(wc_eez_sf, crs(suitable_spots))
crs(wc_eez_sf_matched) == crs(suitable_spots)
cell_area <- cellSize(suitable_spots, unit = "km")
suitable_area <- cell_area * suitable_spots
wc_eez_sf_matched$id <- seq_len(nrow(wc_eez_sf_matched))
eez_raster <- rasterize(wc_eez_sf_matched, suitable_spots, field = "id")
area_in_eez <- zonal(suitable_area, eez_raster, fun="sum", na.rm = TRUE)
names(area_in_eez) <- c("id", "suitable_area_km2")
eez_fin <- wc_eez_sf_matched |>
     mutate(id = row_number()) |>
     left_join(area_in_eez, by = "id") |>
     mutate(suitable_area_km2 = coalesce(suitable_area_km2, 0)) |>
     arrange(desc(suitable_area_km2)) |>
     mutate(rank = row_number())

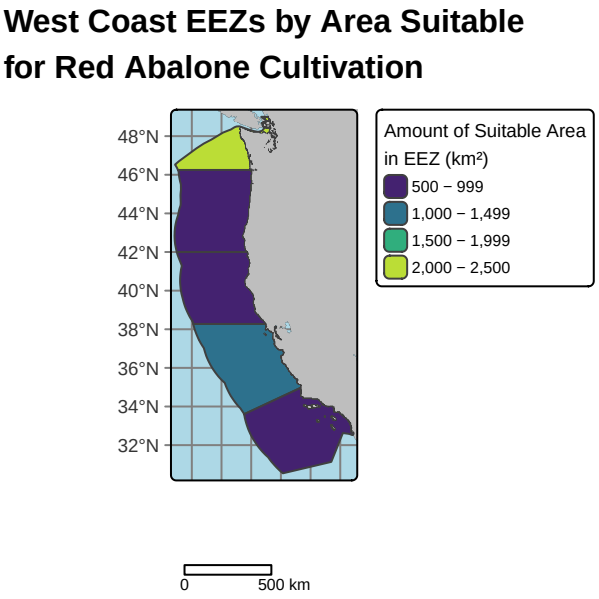
# Map suitability
land <- ifel(depth_resampled > 0, 1, NA)
suit_map <- tm_shape(eez_fin) +
     tm_graticules() +
     tm_polygons("suitable_area_km2", fill.scale =
          tm_scale_intervals(values = "viridis"),
          fill.legend = tm_legend(
               title = "Amount of Suitable Area\nin EEZ (km²)",
               title.size = 0.6,
               text.size = 0.5))  +
     tm_title(
          paste0("West Coast EEZs by Area Suitable\nfor ", species$name, " Cultivation"),
          size = 1,
          fontface = "bold") +
     tm_scalebar(position = tm_pos_out("center", "bottom")) +
     tm_shape(land) +
          tm_raster(col.scale = tm_scale_categorical(values = "grey"),
```

```
            col.legend = tm_legend_hide()) +
          tm_layout(bg.color = "lightblue") +
          tm_options(component.autoscale = FALSE)
    suit_map
}

simplified_aquaculture_analysis(oyster)
```




```
simplified_aquaculture_analysis(red_abalone)
```




```
# This version of the function takes minimum and maximum depth and SST as arugments in additi
aquaculture_analysis <- function(min_sst, max_sst, min_depth, max_depth, species){
     # Read in data
     sst_2008 <- terra::rast(here("data", "average_annual_sst_2008.tif"))
     sst_2009 <- terra::rast(here("data", "average_annual_sst_2009.tif"))
     sst_2010 <- terra::rast(here("data", "average_annual_sst_2010.tif"))
     sst_2011 <- terra::rast(here("data", "average_annual_sst_2011.tif"))
     sst_2012 <- terra::rast(here("data", "average_annual_sst_2012.tif"))
     # Depth file (raster)
     depth <- terra::rast(here("data", "depth.tif"))
     # EEZ file (shapefile)
     wc_eez_sf <- read_sf(here("data", "wc_regions_clean.shp"))

     # Clean/crop/reshape data
     sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
     sst_mean_c <- mean(sst_stack, na.rm=TRUE) - 273.15
     depth_crop <- crop(depth, sst_mean_c)
     depth_proj <- project(depth_crop, sst_mean_c)
     depth_resampled <- resample(depth_proj, sst_mean_c, method = "bilinear")

     # Filter for suitable locations
     depth_matrix <- matrix(c(- Inf, -(max_depth), 0,
                              -(max_depth), -(min_depth), 1,
```

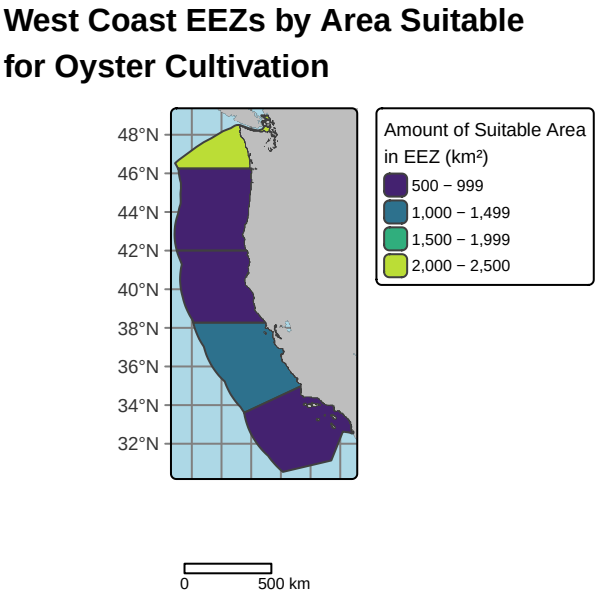
```
                          -(min_depth), Inf, 0),
                          ncol = 3, byrow = TRUE)
depth_reclassified <- classify(depth_resampled, rcl = depth_matrix, include.lowest = TRU
sst_matrix <- matrix(c(-Inf, min_sst, 0,
                          min_sst, max_sst, 1,
                          max_sst, Inf, 0),
                          ncol = 3, byrow = TRUE)
sst_reclassified <- classify(sst_mean_c, rcl = sst_matrix, include.lowest = TRUE)
suitable_spots <- lapp(c(sst_reclassified, depth_reclassified), fun = function(x, y) x*y

# Calculate suitable area by EEZ
wc_eez_sf_matched <- st_transform(wc_eez_sf, crs(suitable_spots))
crs(wc_eez_sf_matched) == crs(suitable_spots)
cell_area <- cellSize(suitable_spots, unit = "km")
suitable_area <- cell_area * suitable_spots
wc_eez_sf_matched$id <- seq_len(nrow(wc_eez_sf_matched))
eez_raster <- rasterize(wc_eez_sf_matched, suitable_spots, field = "id")
area_in_eez <- zonal(suitable_area, eez_raster, fun="sum", na.rm = TRUE)
names(area_in_eez) <- c("id", "suitable_area_km2")
eez_fin <- wc_eez_sf_matched |>
     mutate(id = row_number()) |>
     left_join(area_in_eez, by = "id") |>
     mutate(suitable_area_km2 = coalesce(suitable_area_km2, 0)) |>
     arrange(desc(suitable_area_km2)) |>
     mutate(rank = row_number())

# Map suitability
land <- ifel(depth_resampled > 0, 1, NA)
suit_map <- tm_shape(eez_fin) +
     tm_graticules() +
     tm_polygons("suitable_area_km2", fill.scale =
          tm_scale_intervals(values = "viridis"),
          fill.legend = tm_legend(
               title = "Amount of Suitable Area\nin EEZ (km²)",
               title.size = 0.6,
               text.size = 0.5))  +
     tm_title(
          paste0("West Coast EEZs by Area Suitable\nfor ", species, " Cultivation"),
          size = 1,
          fontface = "bold") +
     tm_scalebar(position = tm_pos_out("center", "bottom")) +
     tm_shape(land) +
```

```
                tm_raster(col.scale = tm_scale_categorical(values = "grey"),
              col.legend = tm_legend_hide()) +
          tm_layout(bg.color = "lightblue") +
          tm_options(component.autoscale = FALSE)
      suit_map
}

aquaculture_analysis(8, 18, 0, 24, "Oyster")
```



Reflection and Intepretation

Based on these graphs, we can see that mainly coastal regions are suitable for oyster cultivation. Additionally, the Central California EEZ contains the most suitable oyster cultivation sites, followed by the Southern California EEZ and the Washington EEZ. In contrast, only the Washington EEZ has a large amount of sites suitable for red abalone cultivation, with some suitable area in the Central California EEZ and almost none in the others.

As shown by these results, our aquaculture_analysis function is a useful tool for preliminary analysis to discover where different commercially useful marine species might best be found or grown. As this tool only takes into account depth and sea surface temperature requirements, it leaves out a great number of other factors that influence species distributions. Therefore,

it likely classifies a number of unsuitable sites as "suitable", and vice versa. Still, it could be useful for preliminary analysis, with mroe comprehensive tools then used to identify sites within the EEZs that our tool identified as most suitable.

Data Sources

Sea surface temperature (SST) data is sourced from NOAA's Daily Global 5km Satellite Sea Surface Temperature Anomaly (https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php).

Depth data comes from GEBCO, General Bathymetric Chart of the Oceans (https://www.gebco.net/data-products/gridded-bathymetry-data#area).

EEZ boundries come from Marineregions.org (https://www.marineregions.org/eez.php).

Information on temperature and depth requirements for marine species come from SeaLifeBase (https://www.sealifebase.ca)